PRODUCT REQUIREMENTS DOCUMENT

Telecom Complaint Intelligence & Automated Resolution Assistant

Professional, customer-first product specification

Cognizant Hackathon • Use Case 13

Document	Specification
Version	7.0 — Autonomous Multi-Task Agent & Multilingual Intelligence PRD
Product Type	AI-powered telecom complaint intelligence and automated resolution platform
Primary User	Telecom customer
Operational User	Support / Network / Operations / Admin teams
Agent Orchestration	LangGraph Multi-Task StateGraph (6-node customer agent & 4-node admin agent)
Multilingual Engine	Hugging Face DistilBERT (multilingual-cased) + Groq Zero-Shot Neural Translation + Rule Fallback
Primary AI Provider	Groq API — Free Plan (Llama-3 models + Whisper voice STT) with complete offline fallback
Voice Stack	Groq Whisper (Multilingual Voice STT) + Browser Web Speech API (Bilingual TTS Readout)
Core Data Store	SQLite (tci.db with WAL mode, foreign keys, and immutable audit logs)
Knowledge Store	ChromaDB persistent vector store + Sentence Transformers (all-MiniLM-L6-v2)
Geolocation Engine	Geoapify Geocoding API (in-memory caching and India coordinate boundary validation)
Core Backend	Python + FastAPI + Uvicorn
UI	Customer Assistant (40/60 Split) + Admin Operations Dashboard + Theme Engine (Dark/Light)

Product vision: Turn telecom complaints from isolated support tickets into a closed-loop intelligence system that can understand the customer in any language, verify the issue with live line diagnostics, resolve eligible problems using grounded SOPs, track unresolved cases transparently, detect mass incidents in real time, and proactively prevent repeated complaints.

This PRD preserves and elevates all capabilities from the original Use Case 13 — complaint classification, sentiment, escalation prediction, RAG, root-cause analysis, heatmaps, spike detection and proactive notification — while structuring them around the customer's complaint-to-resolution journey and modern multi-task agent workflows.

1. Product Overview

The Telecom Complaint Intelligence & Automated Resolution Assistant is a customer-facing AI service backed by an operational support and analytics platform. It accepts complaints through a conversational assistant via text or voice, classifies and prioritizes them, checks for known incidents, retrieves approved troubleshooting guidance, creates and tracks tickets when required, and verifies whether the customer is actually satisfied with the resolution.

The Admin Dashboard is the operations control plane used to manage the complaint lifecycle, investigate patterns, monitor service issues, update ticket states, manage incidents, evaluate AI recommendations, and query operational intelligence.

1.1 Product Principles

- **Customer first:** The primary workflow begins with the customer's problem and ends only when the issue is resolved or properly escalated.
- **One source of truth:** Customer assistant and admin dashboard read/write through the same backend and SQLite database (tci.db).
- **AI with controlled actions:** The LLM interprets and explains; backend APIs perform privileged operations.
- **Dynamic by design:** Complaints can be created by customers, admins or authorized data feeds; the system is not dependent on static CSV data.
- **Closed loop:** A proposed resolution requires customer confirmation before an eligible complaint is closed.
- **Explainable intelligence:** Classification, priority, escalation and root-cause outputs expose useful evidence and contributing factors.
- **Zero-Cost & Offline Resilient:** Runs 100% on free-tier APIs with immediate deterministic offline fallbacks for zero API key operation.

1.2 Problem Statement

Telecom operators receive large volumes of complaints related to network issues, call drops, broadband failures, billing disputes and service requests. Manual triage is slow, customers often lack visibility into what happens after raising a complaint, and support teams may discover mass outages only after complaint volumes increase.

The product must therefore solve two connected problems: **(1) help individual customers reach a verified resolution faster**, and **(2) convert complaint data into operational intelligence** that helps telecom teams detect, investigate and prevent recurring or mass complaints.

2. Objectives

Objective	Expected Outcome
Automate complaint understanding	Classify intent/category, extract key entities, detect sentiment, urgency and escalation risk.
Instant line & speed diagnostics	Provide on-demand telemetry tests (speed, latency, jitter, packet loss) directly inside chat.
Improve first-contact resolution	Use incident checks, ChromaDB RAG and safe troubleshooting workflows before escalating to human support.
Create a complete complaint lifecycle	Support verification, ticket creation, assignment, status updates, resolution, confirmation, reopening and escalation.
Give customers live visibility	Allow authenticated customers to check ticket status and status history from the assistant in natural language.
Reduce duplicate complaints	Match new complaints against active incidents and link reports without creating duplicate tickets.
Detect mass service problems	Use complaint spikes, location and service patterns against rolling baselines to create incident alerts.
Support root-cause investigation	Combine structured complaint patterns with historical/knowledge evidence to generate explainable hypotheses.
Proactively inform customers	Notify affected customers about confirmed/approved incidents and meaningful ticket events.
Operations AI decision support	Provide an autonomous LangGraph agent for NOC admins to query critical issues and SOPs.
Improve operational efficiency	Provide an admin dashboard for queue management, heatmap investigation, analytics and resolution actions.

2.1 Success Criteria

- Customer can report a complaint or run speed diagnostics without navigating complex forms.
- Every valid complaint receives a unique ticket ID (TCK-xxx) and persistent database record.
- Customer can retrieve the current ticket status at any time in English, Hindi, or Hinglish.
- AI-assisted resolution never silently closes a complaint.
- A customer can reject a resolution and reopen/escalate the complaint.
- Admin status changes are reflected in the customer experience in real time.
- Known active incidents can be linked to new complaints to reduce duplicate tickets.
- Admin can see complaint volume, priority, category, sentiment, escalation risk, geographic spikes and incident information.

2.2 Scope Boundaries

- MVP focuses on complaint intelligence and resolution orchestration; it does not directly control telecom network equipment.
- Real OSS/BSS integration is a future integration point; the hackathon uses simulated telemetry APIs and controlled data feeds.
- External SMS/push delivery is optional for MVP; in-app notifications are sufficient to demonstrate the workflow.
- The platform uses the supplied complaint dataset as seed data but does not treat the CSV as the live operational database.

3. Users & Roles

Role	Needs	Key Capabilities
Customer	Fast help, line speed test, clear status, proof that complaint was resolved.	Chat assistant, voice STT/TTS, line diagnostics, ticket creation, troubleshooting, ticket tracking, confirmation, reopen, escalation, notifications.
Support Admin	Efficient queue and resolution management.	Ticket queue, assignment, priority, status updates, notes, resolution actions, SLA monitoring, escalation.
Network Operations	Detect and investigate service-wide problems.	Leaflet heatmap, complaint spikes, incident management, root-cause investigation, affected-customer analysis, broadcast approvals.
Operations AI User	Instant decision support and NOC briefings.	Admin AI Assistant, prompt chips, live database snapshots, ChromaDB SOP synthesis, classification explanations.
System / AI	Automate classification, scoring and decision support.	LangGraph multi-task agents, ML scoring, spike detection, RAG retrieval, response generation, audit logging.

3.1 Access Control

- Customers can access only their own profile, conversations, complaints, notifications and feedback.
- Admins can access operational complaint and incident data according to role.
- AI services do not receive unrestricted database access; they call authorized backend functions.
- All privileged actions are recorded in an audit log.

4. Customer Journey

The customer journey is the central workflow of the product.

Step	Customer Experience	System Behaviour
1. Start	Customer describes issue naturally or runs speed test.	DistilBERT normalizes text; intent router identifies query type.
2. Verify	Customer confirms service/account/issue details.	Authentication + required field validation.
3. Diagnose	Assistant checks known incident and retrieves relevant guidance.	Incident engine + ChromaDB RAG + resolution rules + line diagnostic.
4. Resolve	Customer follows safe troubleshooting steps if eligible.	Resolution attempt recorded.
5. Confirm	Customer says whether service is working.	Confirmation stored as resolution outcome.
6. Ticket	If unresolved, system creates/updates ticket.	Ticket ID, priority (P1-P4), SLA and category persisted in SQLite.
7. Track	Customer asks 'What is the status?'	Current SQLite status + assigned team + history returned.
8. Escalate	Customer requests human help or AI cannot safely resolve.	Ticket routed to support queue with priority escalation.
9. Reopen	Customer says issue returned/not fixed.	Complaint reopened with previous resolution preserved.
10. Close	Customer confirms final resolution.	Complaint moves to CLOSED and 1-5 star CSAT feedback captured.

4.1 Complaint Status Model

NEW → VERIFICATION_REQUIRED → VERIFIED → CLASSIFIED → DIAGNOSING → IN_PROGRESS → RESOLVED_PENDING_CONFIRMATION → CLOSED

Alternative paths: WAITING_FOR_CUSTOMER, HUMAN_ESCALATION, REOPENED. Every status transition creates an immutable status-history record in SQLite with actor, timestamp and reason.

5. AI Customer Assistant

The assistant behaves as a workflow-driven support agent, not as a generic chatbot. It identifies the customer's intent, collects missing information, calls the correct backend function, retrieves relevant knowledge from ChromaDB, and generates a grounded response.

5.1 Supported Intents

Intent	Action
REPORT_COMPLAINT	Collect details → verify → classify → create complaint.
DIAGNOSTIC	Run dynamic network line diagnostic & render speed telemetry card.
CHECK_STATUS	Retrieve authenticated customer's current ticket status/history.
TROUBLESHOOT	Retrieve relevant approved SOP from ChromaDB and guide the customer.
KNOWN_INCIDENT	Check active incident and explain known service impact.
CONFIRM_RESOLUTION	Record customer confirmation and close eligible ticket.
REJECT_RESOLUTION	Move complaint to REOPENED / escalation workflow.
REOPEN_COMPLAINT	Reopen eligible complaint and preserve history.
BILLING_QUERY	Provide approved billing guidance or route to support.
ESCALATE	Create human-support escalation with priority adjustment.
GENERAL_QUERY	Answer from approved knowledge or ask clarification.

5.2 Assistant Architecture (LangGraph 6-Node Workflow)

Customer message → **1. translate_input** (DistilBERT) → **2. route_intent** (Intent Classifier) → **3. retrieve_context** (ChromaDB RAG + Incidents) → **4. execute_action** (Speed Test / DB Mutations) → **5. synthesize_response** (Grounded Groq LLM) → **6. translate_output** (Devanagari Script & Cards).

5.3 Important Design Rule

The LLM does not directly change ticket status or invent live data. It can interpret the request and generate language, but actions such as creating a complaint, checking status, changing status, confirming resolution and reopening a complaint are performed through authenticated backend APIs.

6. Solution Architecture

Layer	Components	Purpose
Customer Layer	React 40/60 chat, voice STT/TTS, diagnostic cards	Complaint reporting, speed diagnostics, troubleshooting, tracking and confirmation.
Admin Layer	React operations dashboard, AI Assistant, Theme Engine	Queue, ticket management, incidents, heatmap, analytics, alert inbox, notify queue, audit.
API Layer	FastAPI + JWT authentication + scheduler	Central business logic, authorization and deterministic actions.
AI Orchestration	LangGraph Multi-Task StateGraph	Controls multi-turn state continuation and tool execution.
Multilingual NLP	Hugging Face DistilBERT (multilingual-cased)	Tokenization, Hinglish normalization, Devanagari validation.
ML Intelligence	scikit-learn (TF-IDF + LogReg), Multi-Factor Scoring	Classification (Macro-F1 >= 82.4%), sentiment, urgency, escalation, priority scoring.
Vector RAG	ChromaDB + Sentence Transformers (all-MiniLM-L6-v2)	Grounded troubleshooting, FAQs, SOPs and approved resolved cases.
Generative AI	Groq API (Llama-3, Whisper) + Offline Fallbacks	Conversation generation, summaries, explanations and root-cause narrative.
Geolocation	Geoapify Geocoding API + dynamic cache	Location coordinates for Leaflet regional heatmap.
Operational DB	SQLite (tci.db with WAL mode)	Live complaints, users, status history, incidents, notifications and audit logs.
Analytics / Detection	Spike detection, heatmap, incident engine	Proactive service intelligence.

6.1 Data Flow

Customer/Admin/API input → validation → complaint persistence → SQLite → incident/duplicate evaluation → resolution/ChromaDB RAG → ticket lifecycle → notification/analytics → feedback.

SQLite is the operational source of truth. ChromaDB is the knowledge retrieval layer. Groq is the generation/reasoning service. This separation prevents the chatbot from becoming the database or the source of live ticket truth.

7. Admin Operations Dashboard

All useful admin-dashboard capabilities from the original concept are retained and organized into 16 operational modules.

Dashboard Module	Features & Operational Capabilities
1. Overview	Total complaints, open tickets, resolved/closed tickets, high-priority complaints, SLA breaches, active incidents, complaint trends.
2. Complaint Queue	Search/filter by status, category, service, region, priority, sentiment, escalation risk, date and incident.
3. Ticket Management	View ticket detail, customer-safe summary, category, priority, AI analysis, incident link, status, SLA, assignment and history.
4. Assignment & Escalation	Assign to support team/person (Field Ops, RF Team, Billing, Support L2, Network Ops), mark waiting for customer.
5. Resolution Management	Add resolution notes, propose resolution, record resolution source, move to RESOLVED_PENDING_CONFIRMATION.
6. Status Management	NEW, IN_PROGRESS, WAITING_FOR_CUSTOMER, ESCALATED, RESOLVED_PENDING_CONFIRMATION, CLOSED, REOPENED.
7. Complaint Intelligence	Category distribution, sentiment, urgency, escalation risk, priority distribution and recurring issue analysis.
8. Heatmap	Leaflet geographic complaint density with filters and drill-down to underlying complaints (Red=Spike, Amber=Elevated, Teal=Normal).
9. Spike Detection	Rolling baseline comparison, abnormal complaint volume alerts and automatic incident creation.
10. Root-Cause Investigator	Likely cause, confidence bar (90%+), evidence signals (>=3 bullets), affected region/service and supporting complaint patterns.
11. Incident Management	Create/acknowledge/assign/update/resolve incidents; link complaints; inspect affected customers.
12. Proactive Alerts	Admin alert inbox for complaint spikes, SLA risks, active incidents and escalation risks with unread badges.
13. Notification Center	Draft, review and approve incident/customer notifications; view delivery/read state for in-app notifications.
14. Executive Analytics	Resolution rate, response time, escalation rate, SLA performance, category trends, duplicate complaints and customer feedback.
15. Data Ingestion	Admin-only 3-step CSV upload wizard with auto-schema detection, PII redaction and pipeline triggering.
16. Operations AI Assistant	LangGraph decision support agent answering complex operational queries with live DB snapshots and ChromaDB SOPs.

7.1 Dashboard Principle

The dashboard helps an operator answer four questions quickly: **What is happening? Which customers are affected? What should we do? Has the customer actually been helped?**

8. Complaint Intelligence Pipeline

Capability	Input	Output
Text Classification	Complaint text	Category/subcategory/service type (Macro-F1 >= 82.4%).
Intent Detection	Conversation message	Supported assistant intent (15+ categories).
Entity Extraction	Complaint text	Location, service, device, timing and issue details.
Sentiment Analysis	Complaint text/conversation	Positive/neutral/negative/critical signal.
Urgency Detection	Complaint + context	Urgency score/level (0.0 to 1.0).
Escalation Risk	Complaint + history + sentiment	Escalation probability/risk level (churn & TRAI threats).
Priority Scoring	Urgency + impact + risk + incident context	P1/P2/P3/P4 priority with contributing factor chips.
Duplicate Detection	New complaint + existing complaints/incidents	Similarity/link recommendation.
Ticket Summary	Full conversation/complaint	Concise agent-ready 1-sentence summary.

8.1 Priority Example & Formula

Priority = clamp(0, 100, w1*Urgency + w2*Sentiment + w3*Escalation + w4*Incident + w5*Repeat)

- **P1 (Critical, 80-100):** SLA 2 Hours — High priority alert drafted; immediate queue banner.
- **P2 (High, 60-79):** SLA 6 Hours — Routed to Tier-2 specialist queue.
- **P3 (Medium, 35-59):** SLA 24 Hours — Standard operational queue.
- **P4 (Low, 0-34):** SLA 48 Hours — General queue.

The dashboard displays the major contributing factor chips rather than only a numeric score.

9. RAG-Based Automated Resolution

RAG is used to make the assistant useful for actual resolution rather than only classification.

9.1 Knowledge Base

- Telecom troubleshooting SOPs (broadband reset, ONT reconfiguration, APN settings, eSIM).
- FAQs and approved service guidance.
- Billing/service policies.
- Historical confirmed resolutions.
- Incident resolution notes and post-mortems.

9.2 Resolution Decision

Condition	Assistant Behaviour
Known incident exists	Explain incident, provide available guidance and link complaint to incident where appropriate.
Known low-risk issue + good RAG match	Guide customer through safe step-by-step troubleshooting SOPs.
Insufficient knowledge	Ask clarification or escalate; do not hallucinate.
Account-sensitive issue	Use authenticated backend data or route to human support.
Troubleshooting succeeds	Ask customer to confirm resolution (Yes ✓).
Customer rejects resolution	Reopen/escalate and preserve the previous attempt.

9.3 RAG Grounding Rule

Every troubleshooting response should be generated from retrieved approved knowledge. The assistant should not claim a cause, ETA, policy or fix that is not supported by the retrieved ChromaDB chunks or live SQLite backend state.

10. Dynamic Data & Complaint Lifecycle

The original dataset remains useful for model training/evaluation and demonstration, but runtime operation is database-driven.

Data Source	Use
Historical CSV / Demo data	Model development, evaluation, historical analytics and seed records.
Customer Assistant	Real-time complaint creation and customer interactions.
Admin Dashboard	Status updates, assignments, resolution notes and incident actions.
Dynamic Line Diagnostics	Real-time simulated telemetry for line speed, latency, jitter and packet loss.
Background Detection	Complaint aggregation, spike detection and incident generation.

10.1 Core Database Entities

Entity	Purpose
users	Customer/admin identity and role.
complaints	Live complaint/ticket record with priority and SLA.
complaint_status_history	Immutable status transition history.
chat_messages	Conversation messages, detected intents and metadata.
resolutions	Troubleshooting/resolution attempts and confirmation.
incidents	Mass-service issue and root-cause information.
notifications	Ticket/incident notification records.
kb_docs	RAG source metadata.
feedback	Customer resolution rating (1-5 stars) and comments.
audit_logs	Operational/security trace.

10.2 Required Complaint Fields: complaint_id, customer_id, text, category, region, lat, long, service_type, sentiment, urgency, escalation_risk, priority_score, priority_label, sla_deadline, status, incident_id, assigned_to, ticket_summary, created_at

11. Verification, Status & Resolution Management

This section addresses the most important gap in a basic chatbot: what happens after the customer reports the issue?

11.1 Customer Verification

- Authenticate customer before exposing personal complaint information.
- For a new complaint, summarize extracted details and ask for confirmation before ticket creation.
- For an existing complaint, identify it through the authenticated customer's account rather than trusting an arbitrary ticket ID.
- If required information is missing or inconsistent, ask targeted clarification questions.

11.2 Ticket Status Visibility

- Customer can ask for the status in natural language (English, Hindi, Hinglish).
- Assistant retrieves the current authoritative state directly from SQLite (tci.db).
- Response includes status, last update, assigned team, SLA countdown, and a short explanation.
- Status history is available for transparency in the customer drawer.

11.3 Resolution Confirmation

- AI/admin proposes a resolution → Ticket enters RESOLVED_PENDING_CONFIRMATION.
- Customer is explicitly asked whether the issue is fixed.
- **YES** → record confirmation → CLOSED → CSAT feedback.
- **NO** → REOPENED / HUMAN_ESCALATION → support queue.

11.4 Customer Feedback

After closure, the customer can provide a 1–5 star rating and optional comment. Feedback is stored against the complaint and surfaced in admin analytics for resolution-quality monitoring.

12. Mass Complaint Intelligence & Root-Cause Analysis

The operational intelligence features turn raw complaints into actionable network insights.

12.1 Complaint Spike Detection

- Group complaints by time window (6h rolling), region, category and service type.
- Compare current volume against a rolling 7-day historical baseline.
- Flag statistically abnormal increases (3x to 300x surge) according to configured thresholds.
- Create or update an incident record and link future matching complaints.

12.2 Geographic Heatmap

- Show complaint density by region via interactive Leaflet map.
- Filter by time, category, service and severity.
- Drill down from region → incident → complaint list.
- Highlight abnormal regions (Red surge circles) and active incidents.

12.3 AI Root-Cause Investigator

Evidence Signal	Analytical Computation	Example
Volume anomaly	Multiplier vs historical rolling baseline.	Complaint count is 352x above baseline in Raj Nagar.
Geographic concentration	Percentage of complaints in target area.	94.2% of complaints are from Ghaziabad circle.
Service concentration	Dominant service proportion.	98.1% of complaints involve broadband/fiber.
Time correlation	Temporal cluster window onset.	Complaints started sharply within a 2-hour window.
Historical similarity	Cosine similarity to past incidents.	92% match to INC-2025-0873 (Optical fiber cut).
Knowledge evidence	ChromaDB SOP corroboration.	Retrieved SOP supports physical line severance.

Root-cause output: likely cause + confidence gauge (90%+) + evidence signals (>=3 checkable bullets) + affected scope + recommended action. AI output remains a hypothesis until confirmed by operations.

13. Proactive Customer Notification

Trigger	Notification Content & Purpose
Ticket created	Complaint/ticket reference ID, summary and SLA deadline target.
Ticket assigned	Support ownership update indicating assigned engineering department.
Status changed	Meaningful lifecycle update with customer-visible admin note.
Resolution proposed	Customer asked to verify service in chat before closure.
Incident detected/confirmed	Affected customers in region informed of known issue and ETA.
Complaint reopened	Customer and support receive relevant update on escalation.
SLA risk/breach	Customer and support alerted where priority threshold is reached.

13.1 MVP Notification Policy

In-app transactional updates are generated automatically from authenticated complaint state. **Mass proactive incident notifications must be reviewed and approved by an admin in the Notify Queue (/admin/notifications) to eliminate false-positive broadcast spam.** SMS/push provider integration is supported as optional webhooks.

14. Dynamic Network & Line Diagnostics

To deliver real-time utility beyond conversational text, the platform provides an on-demand network telemetry diagnostic tool (/api/chat/diagnostic).

Metric	Broadband / Fiber	Mobile Data	Degraded (Active Outage)
Download Speed	94.0 - 298.0 Mbps	35.0 - 78.0 Mbps	1.5 - 12.0 Mbps
Upload Speed	88.0 - 290.0 Mbps	15.0 - 32.0 Mbps	0.5 - 4.0 Mbps
Ping Latency	12.0 - 28.0 ms	22.0 - 45.0 ms	120.0 - 280.0 ms
Jitter	1.2 - 4.5 ms	3.5 - 8.0 ms	25.0 - 65.0 ms
Packet Loss	0.0%	0.0 - 0.5%	8.0 - 22.0%
Line Health	Optimal / Healthy	Optimal / Healthy	Degraded (Incident Linked)

14.1 Diagnostic Workflow

Customer requests speed test → Backend inspects customer region & active incidents → Returns telemetry payload → Rendered as visual diagnostic card in chat → Read out via bilingual TTS audio.

15. Backend REST API Specifications

Method & Route	Role	Purpose & Functionality
POST /api/auth/login	Public	Customer/admin authentication; returns signed JWT.
POST /api/auth/signup	Public	Customer self-registration with region and service.
POST /api/chat	Customer	Conversational endpoint executing LangGraph StateGraph.
POST /api/chat/diagnostic	Customer	Executes real-time network speed/latency diagnostics.
POST /api/chat/voice	Customer	Multilingual voice transcription via Groq Whisper.
GET /api/my/tickets	Customer	Lists authenticated customer's own tickets.
GET /api/my/tickets/{id}/history	Customer	Status history timeline for a specific owned ticket.
POST /api/admin/upload/ingest	Admin	Universal CSV ingestion with auto-schema mapping & ETL.
GET /api/admin/queue	Admin	Filterable, paginated priority-ranked ticket queue.
PATCH /api/admin/complaints/{id}	Admin	Updates ticket status, team assignment, and notes.
POST /api/admin/complaints/{id}/propose-resolution	Admin	Proposes technical fix (resolved_pending_confirmation).
GET /api/admin/heatmap	Admin	Regional complaint density & spike data for Leaflet map.
GET /api/admin/incidents	Admin	Lists active outage incidents & root-cause dossiers.
POST /api/admin/assistant/chat	Admin	Admin AI Assistant decision support LangGraph endpoint.
GET /api/admin/audit	Admin	Immutable security & administrative audit logs.

15.1 API Security

• Signed JWT authentication on all protected routes.
• Customer ownership verification on every complaint read/write.
• Server-side validation of all status transitions.
• Comprehensive audit logging for privileged actions.
• API keys and secrets stored securely in environment variables.

16. Free / Low-Cost AI & API Strategy

The platform is engineered to run on 100% free-tier or open-source components with zero required software spend.

Component	MVP Choice	Operational Role & Cost Profile
Generative LLM API	Groq Free Plan	Assistant responses, ticket summaries, root-cause narratives (Zero cost).
Speech Recognition	Groq Whisper	Multilingual voice complaint transcription (Zero cost).
Embeddings	sentence-transformers	all-MiniLM-L6-v2 local CPU embeddings (Zero cost).
Vector Store	ChromaDB (local)	Persistent vector store for SOP retrieval (Zero cost).
Multilingual Tokenizer	Hugging Face DistilBERT	distilbert-base-multilingual-cased (Zero cost).
Classification & ML	scikit-learn	Complaint category and intent classification (Zero cost).
Database	SQLite (local)	Live operational state in WAL mode (Zero cost).
Backend	Python + FastAPI	REST API and background scheduler (Zero cost).
Geocoding	Geoapify (Free tier)	Location coordinates with memory cache (Zero cost).

16.1 Deterministic Offline Fallbacks & Cost Statement

Target is $0 software/API spend for the hackathon demo. Every external API call includes an immediate deterministic offline fallback, guaranteeing that the entire application operates and all automated tests pass even with no API keys or internet connection.

17. Non-Functional Requirements

Category	Requirement & Implemented Standard
Performance	Customer/API interaction latency <= 1.5s on live Groq LLM, <= 50ms on offline fallback.
Availability	Ticket/status operations remain 100% usable even if external LLM is offline.
Security	Scrypt password hashing, signed JWT tokens, role-based authorization guards.
Privacy	Automatic regex PII redaction of names, phone numbers, and emails upon ingestion.
Reliability	Deterministic SQLite database transactions; AI failures never create invalid states.
Explainability	AI classification, priority scores, and root causes expose contributing factors and evidence.
Auditability	All status, assignment, resolution, escalation and notification actions are logged immutably.
Language	Full bidirectional English, Hindi (Devanagari), and Hinglish transliteration support.
Cost	100% free-tier architecture with zero cloud infrastructure overhead.

17.1 AI Safety Rules

- Never fabricate live ticket status or ticket IDs.
- Never claim an outage is confirmed when it is only an unverified AI hypothesis.
- Never claim resolution without explicit customer confirmation.
- Never reveal another customer's private data.
- Never perform privileged database actions directly from generated text.
- When confidence or evidence is insufficient, ask clarification or escalate.

18. Recommended Technology Stack

Layer	Technology Choice & Version
Customer UI	React 18/19, Vite, 40/60 Split Layout, Theme Engine (Dark/Light mode).
Admin Dashboard	React + Leaflet + React-Leaflet + Recharts + Admin AI Assistant.
Backend	Python 3.10+ + FastAPI + Uvicorn.
Database	SQLite 3 with WAL Mode (tci.db).
Vector Store	ChromaDB Persistent Client (backend/data/chroma_db).
Embeddings	Sentence Transformers (all-MiniLM-L6-v2).
Multilingual NLP	Hugging Face Transformers (distilbert-base-multilingual-cased).
LLM & Speech	Groq Cloud API (Llama-3.3-70b, Llama-3.1-8b, Whisper-large-v3-turbo).
Geocoding	Geoapify Geocoding REST API (Cached).

18.1 Separation of Responsibilities

System	Should Do	Should Not Do
SQLite	Store authoritative live operational state.	Generate natural-language answers.
ChromaDB	Retrieve semantic SOP knowledge.	Store authoritative live ticket status.
scikit-learn ML	Classify/score structured complaint data.	Perform privileged database mutation.
Groq LLM	Reason over supplied context and generate language.	Invent current system state or ticket IDs.
FastAPI	Authorize and execute state transitions.	Depend on LLM text for authorization.
Admin Dashboard	Operate, monitor, and approve broadcasts.	Bypass backend authorization or audit logs.

19. Testing & Acceptance Criteria

Scenario	Acceptance Criteria
New complaint	Customer describes issue → assistant collects details → customer verifies → ticket created.
Known outage	Assistant identifies matching incident → links complaint → avoids unnecessary duplicate ticket.
RAG troubleshooting	Assistant retrieves approved SOP → customer performs steps → confirmation requested.
Unresolved issue	Customer rejects fix → complaint reopens/escalates → support queue receives ticket.
Status check	Customer asks status → current DB state returned, not stale model memory.
Admin update	Admin changes status → customer sees updated state on next request/notification.
Spike detection	Synthetic complaint burst → abnormality detected → incident appears on dashboard.
Root cause	Dashboard shows hypothesis with evidence bullets and confidence gauge (90%+).
Authorization	Customer cannot read another customer's complaint even with its ticket ID.
AI outage	Groq unavailable → deterministic status/ticket APIs still work with offline fallbacks.

19.1 Key KPIs & Validated Benchmarks

- **Complaint classification macro-F1:** >= 82.4% (Exceeds >= 80% target on held-out split).
- **Intent routing accuracy:** >= 95.0% across 15+ supported test intents.
- **Ticket creation correctness:** 100% in automated workflow tests.
- **Status retrieval correctness:** 100% in authorized customer tests.
- **Resolution confirmation capture:** 100% of AI-assisted resolution attempts.
- **Test suite pass rate:** 100% (106+ passing tests in backend/tests/).

20. Implementation Plan & Demo

Phase	Implementation Milestone
1	SQLite schema + authentication + role-based access + demo accounts.
2	FastAPI complaint/status APIs + universal CSV schema-mapper + audit logging.
3	LangGraph customer assistant + conversation state + intent routing + DistilBERT.
4	Complaint verification + ticket creation + live status tracking + line speed diagnostic.
5	ChromaDB vector store + SentenceTransformers + SOP troubleshooting workflow.
6	Resolution confirmation + confirm-to-close loop + reopen + escalation.
7	Admin dashboard queue + ticket drawer + team assignment.
8	Leaflet heatmap + Geoapify geocoding + spike detection + incident management.
9	Root-cause investigator + evidence dossier panel + confidence gauge.
10	Proactive notification queue + admin approval + in-app notification feed.
11	LangGraph Admin AI Assistant + prompt chips + decision support cockpit.
12	Theme engine (Dark/Light), multilingual voice STT/TTS, full test suite validation.

20.1 Minimum Successful Demo

- Customer reports a broadband/network problem in natural language (English/Hindi/Hinglish).
- Assistant verifies details and checks whether the region has an active incident.
- Known issue is handled through incident explanation or ChromaDB RAG troubleshooting.
- Unresolved complaint becomes a ticket with P1-P4 priority score and SLA countdown.
- Admin updates the ticket; customer sees the update in real time.
- Customer rejects an attempted resolution; ticket reopens/escalates.
- Outage spike appears on the Leaflet heatmap in red.
- System creates an incident and generates an evidence-backed root-cause dossier.
- Admin reviews/approves a proactive broadcast customer notification.
- Admin queries the Operations AI Assistant for decision support.

21. Product Differentiation

Traditional Complaint System	Proposed TelConnect Platform
Collects complaint and creates passive ticket.	Understands complaint and attempts autonomous grounded resolution.
Customer waits for support.	Assistant provides immediate line diagnostics & grounded SOP troubleshooting.
Ticket status is opaque.	Customer has live status, assigned team, SLA countdown and full history.
Agent marks resolved unilaterally.	Customer explicitly confirms resolution (confirm-to-close loop); rejected fixes reopen.
CSV/dataset is primary data source.	Database is dynamic SQLite; CSV is only a seed/analysis source.
Dashboard is mainly passive reporting.	Dashboard is an operational control plane with LangGraph AI Assistant.
Outages discovered after many calls.	Complaint spikes trigger real-time anomaly alerts & root-cause dossiers.
Generic AI chatbot.	Controlled LangGraph multi-task orchestrator connected to real APIs.

21.1 Final Product Definition

A customer-first AI complaint resolution platform that combines conversational support, dynamic network diagnostics, automated troubleshooting, live ticket management, customer verification and resolution confirmation with telecom operational intelligence such as classification, sentiment, escalation prediction, duplicate detection, geographic heatmaps, complaint spike detection, AI-assisted root-cause analysis and proactive incident notification.

22. Recommended Presentation Flow

Scene	Demonstration	Message to Judges
Customer problem	Customer reports telecom issue in Hindi/Hinglish.	AI understands intent in any language.
Line diagnostic	Customer triggers on-demand speed test.	Real-time network telemetry inside chat.
Intelligence	Assistant detects category, sentiment, urgency.	Complaint intelligence happens immediately.
Resolution	RAG provides troubleshooting or incident info.	AI tries to solve before escalating.
Ticket	Unresolved case becomes structured P1-P4 ticket.	No complaint is lost.
Tracking	Customer asks for status later.	Live dynamic backend single source of truth.
Confirmation	Customer accepts (Yes ✓) or rejects fix.	Closed-loop customer experience.
Admin Cockpit	Support team sees and operates the same ticket.	Zero data drift across surfaces.
Mass issue	Complaint spike appears on Leaflet heatmap.	System sees network patterns.
Root cause	AI provides evidence-backed dossier (92%).	Actionable explainable intelligence.
Proactive alert	Notification is approved for affected users.	Human-in-the-loop reduces duplicates.
Ops AI Assistant	Admin queries NOC operations agent.	Executive decision support cockpit.

23. Requirements Traceability to Use Case 13

Use Case Requirement	PRD Implementation Details
Classify complaint categories	TF-IDF + Logistic Regression (Macro-F1 >= 82.4%).
Detect customer sentiment	Sentiment Analysis & negative urgency weighting.
Prioritize critical complaints	Multi-Factor Priority Scoring + Admin Queue + SLA deadlines.
Predict escalation risk	Escalation Risk model (churn & regulatory threats) + alerts.
Recommend resolution actions	ChromaDB Vector RAG + approved SOP procedures.
Generate ticket summaries	Groq LLM / template fallback summary stored with complaint.
Complaint triage assistant	LangGraph 6-node Customer Multi-Task StateGraph.
Vector DB / RAG	ChromaDB persistent vector store + SentenceTransformers.
Root-cause / intelligence	Spike detection + Leaflet heatmap + Root-Cause Investigator.
Operations AI decision support	LangGraph Admin Operations Agent (/admin/assistant).
Customer experience	40/60 Split chat, line diagnostics, confirm-to-close loop.

24. Conclusion & Appendix

This PRD defines a complete, professional product rather than a collection of disconnected AI features. The customer assistant understands and resolves individual problems; the shared SQLite backend guarantees live complaint state; the admin dashboard manages operations; and the intelligence layer turns complaint patterns into proactive service insights.

The strongest demonstration of the product is therefore not 'the chatbot can answer a complaint.' It is: 'The system understands the complaint, tries to resolve it, creates and tracks a real ticket when needed, verifies the customer's resolution, and simultaneously learns from complaint patterns to detect and prevent larger telecom problems.'

Appendix — Provider / API Note: Groq Cloud API operates within published free-tier rate limits with OpenAI-compatible endpoints. All features include deterministic offline fallbacks ensuring 100% resilience with zero API key dependencies.